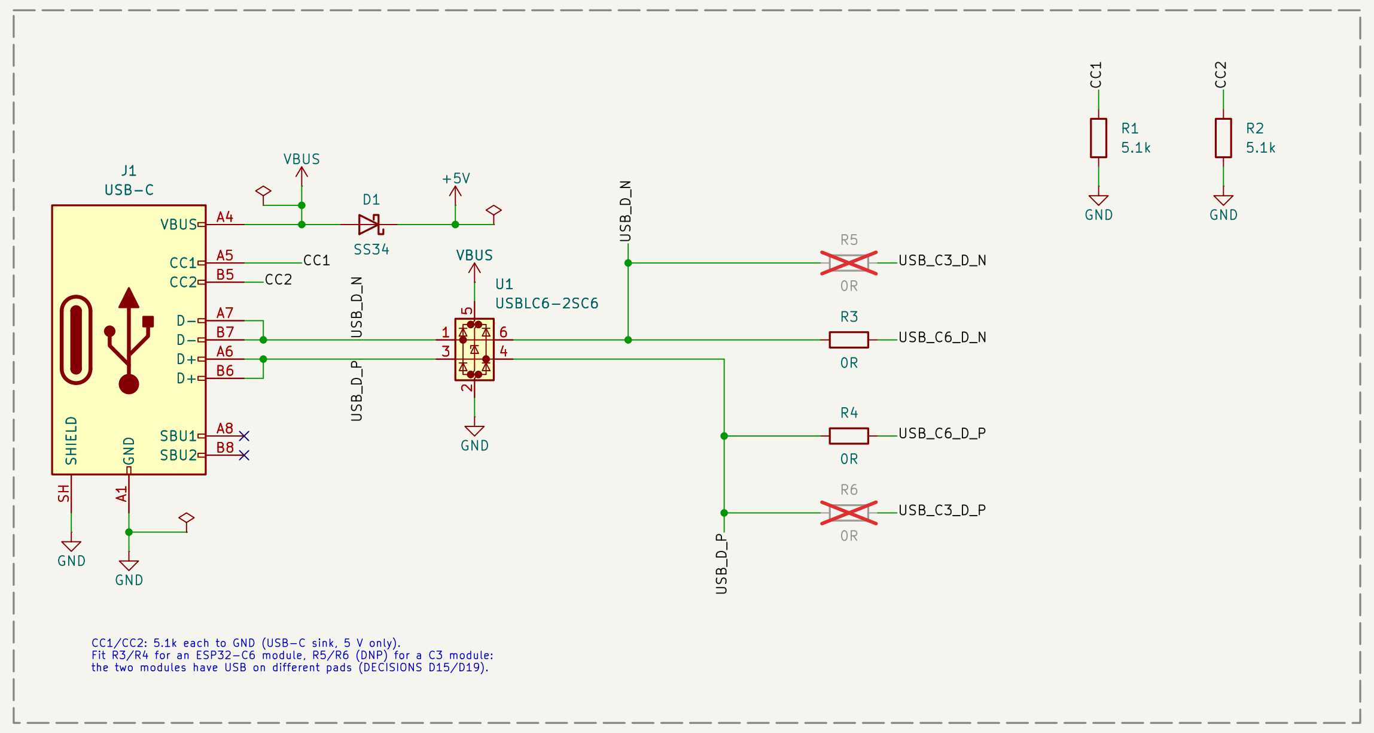
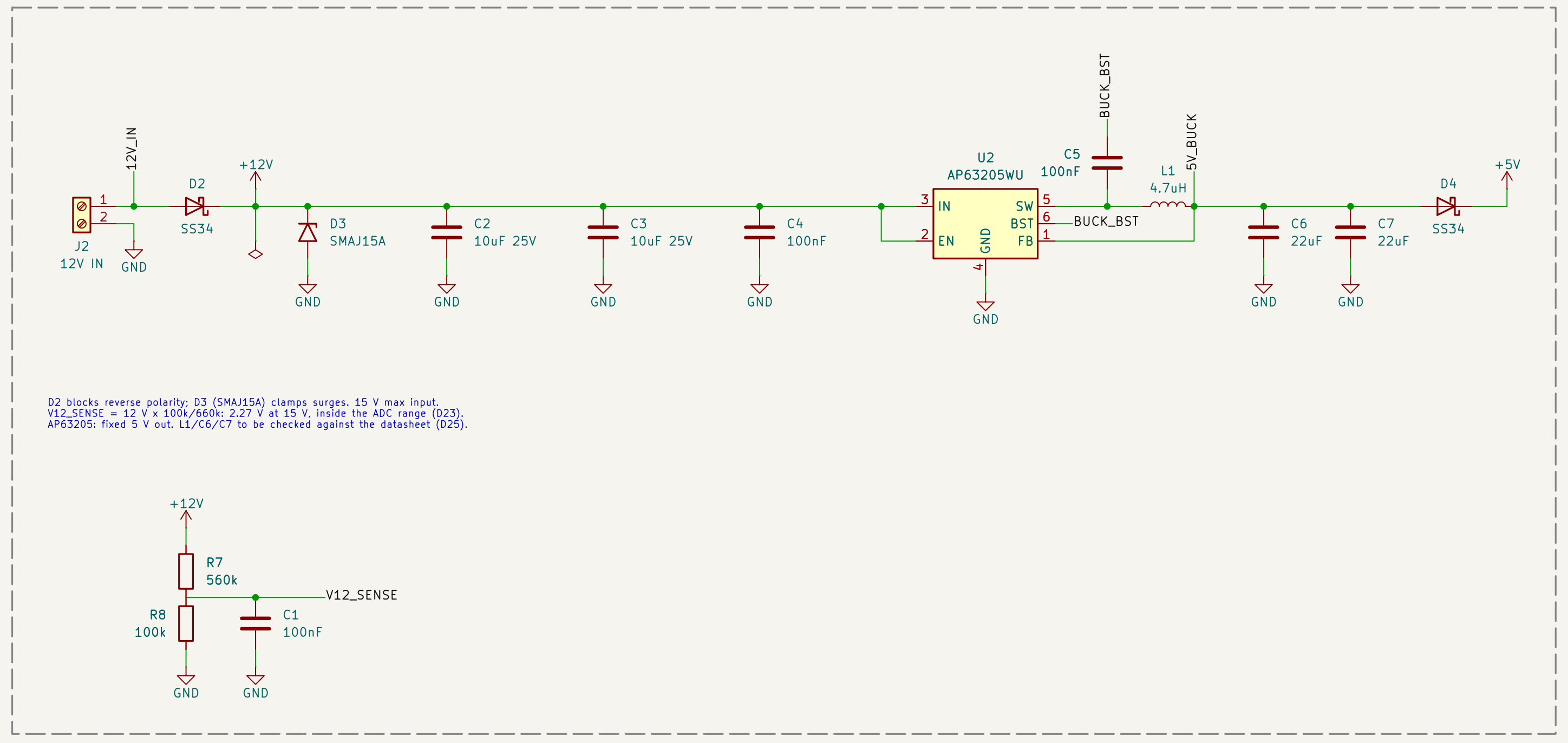
USB-C INPUT



CC1/CC2: 5.1k each to GND (USB-C sink, 5 V only).
Fit R3/R4 for an ESP32-C6 module, R5/R6 (DNP) for a C3 module:
the two modules have USB on different pads (DECISIONS D15/D19).

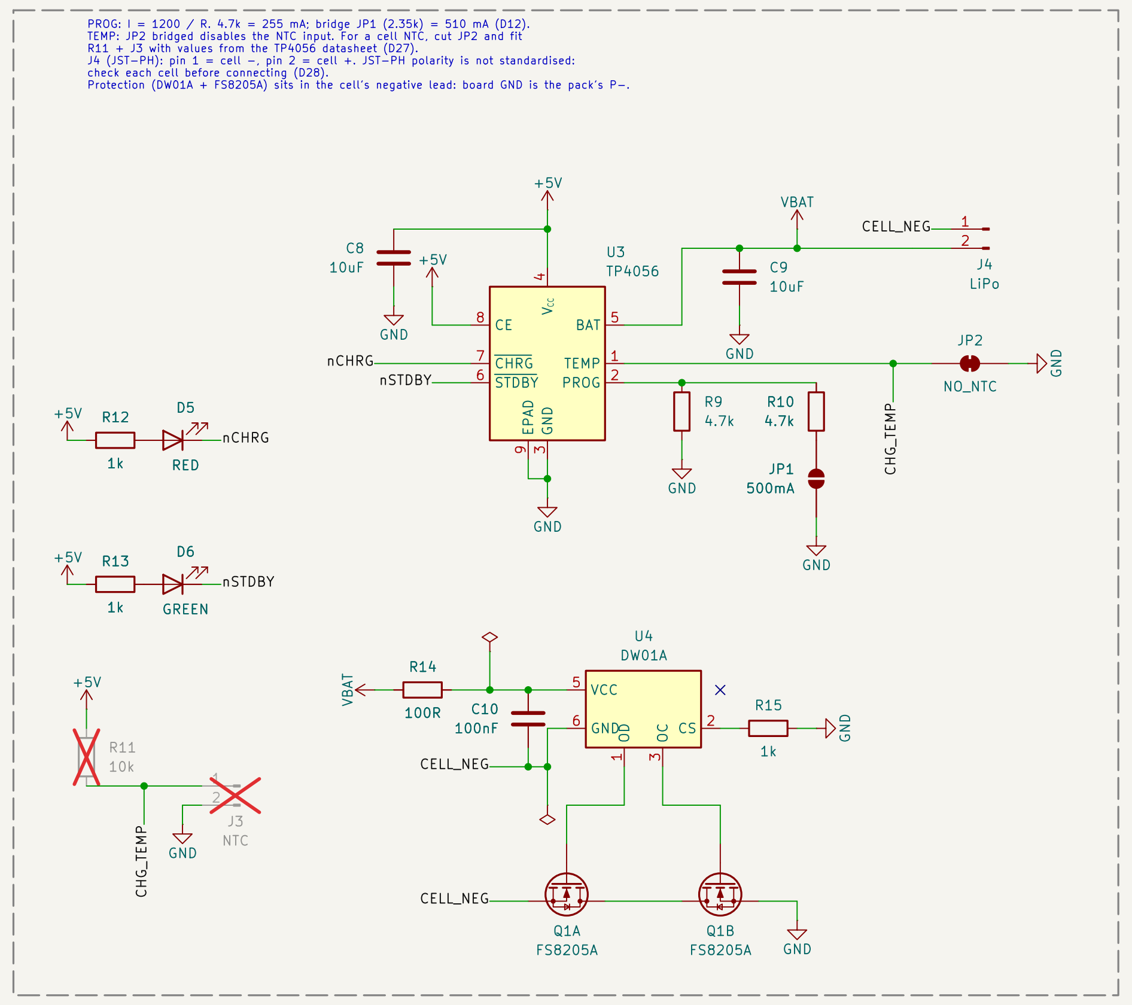
12 V INPUT, PROTECTION & 5 V BUCK



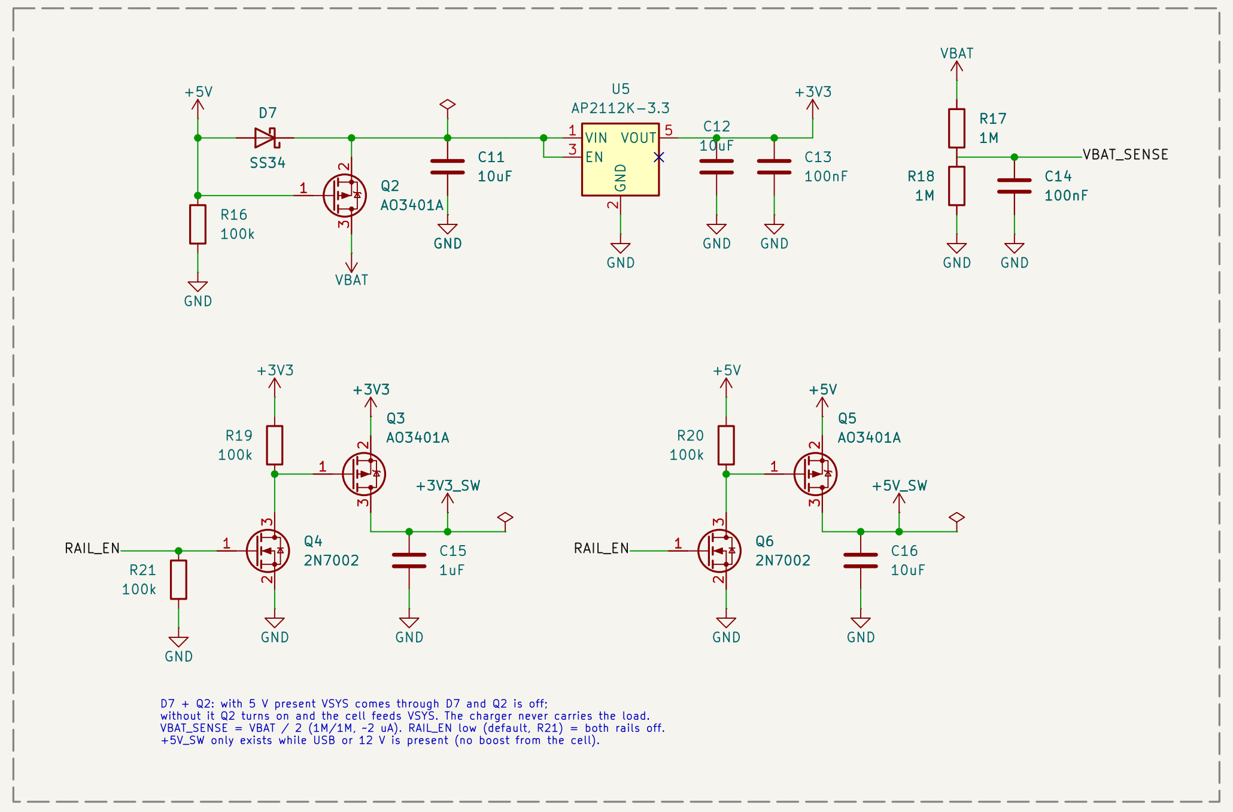
D2 blocks reverse polarity; D3 (SMAJ15A) clamps surges. 15 V max input.
V12_SENSE = 12 V x 100k/660k: 2.27 V at 15 V, inside the ADC range (D23).
AP63205: fixed 5 V out. L1/C6/C7 to be checked against the datasheet (D25).

LI-ION CHARGER & CELL PROTECTION

PROG: I = 1200 / R. 4.7k = 255 mA; bridge JP1 (2.35k) = 510 mA (D12).
TEMP: JP2 bridged disables the NTC input. For a cell NTC, cut JP2 and fit
R11 + J3 with values from the TP4056 datasheet (D27).
J4 (JST-PH): pin 1 = cell −, pin 2 = cell +. JST-PH polarity is not standardised:
check each cell before connecting (D28).
Protection (DW01A + FS8205A) sits in the cell's negative lead: board GND is the pack's P−.

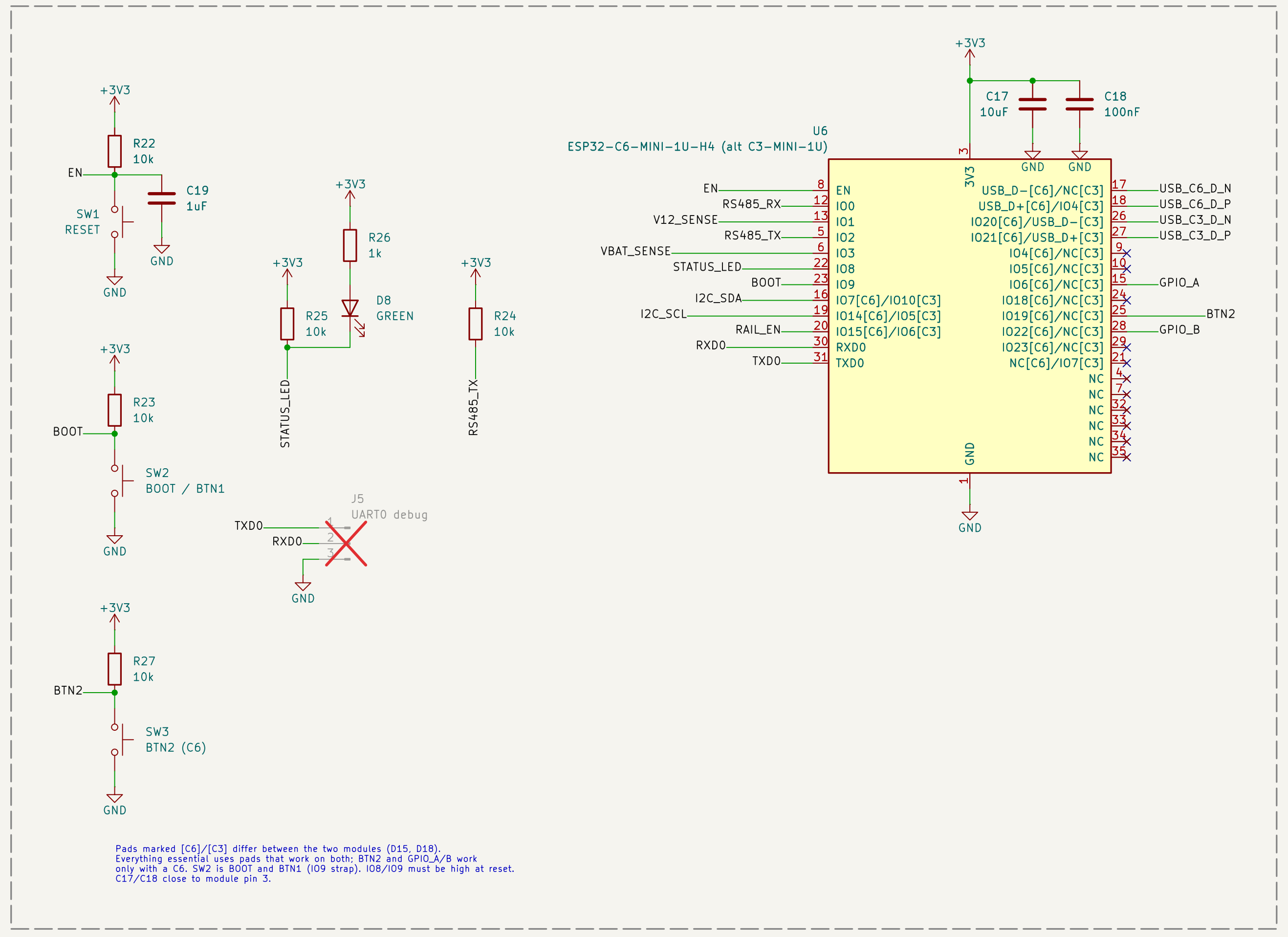


SYSTEM POWER: LOAD SHARING, 3.3 V LDO, SWITCHED RAILS



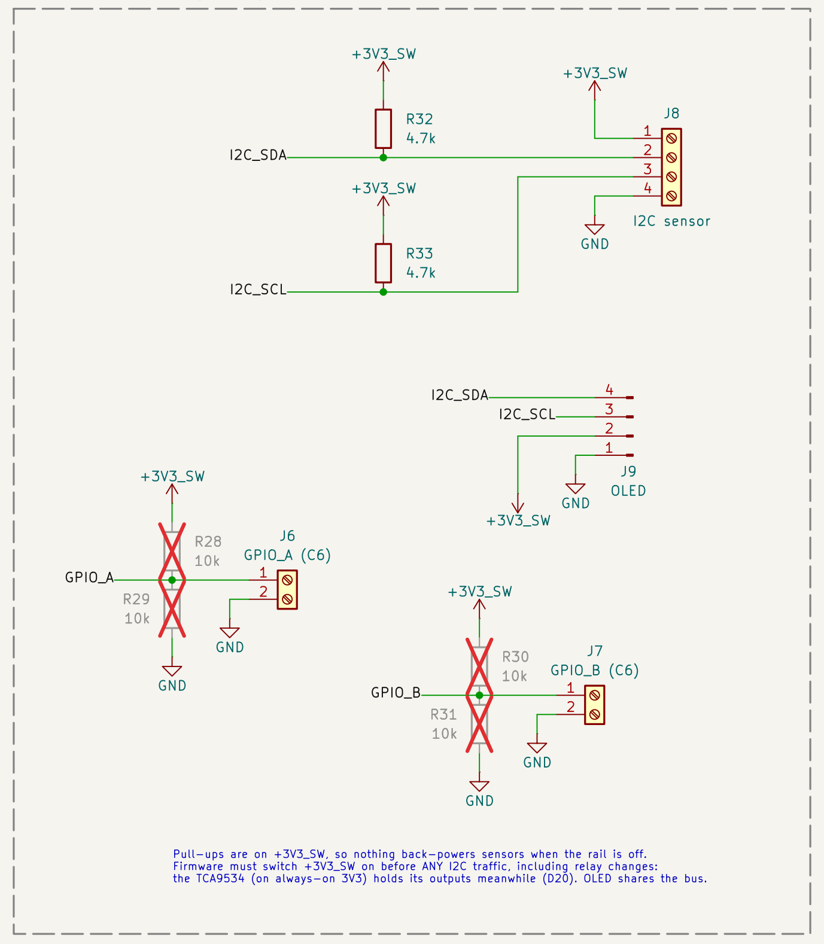
D7 + Q2: with 5 V present VSYS comes through D7 and Q2 is off;
without it Q2 turns on and the cell feeds VSYS. The charger never carries the load.
VBAT_SENSE = VBAT / 2 (1M/1M, ~2 uA). RAIL_EN low (default, R21) = both rails off.
+5V_SW only exists while USB or 12 V is present (no boost from the cell).

ESP32-C6 / C3 MODULE



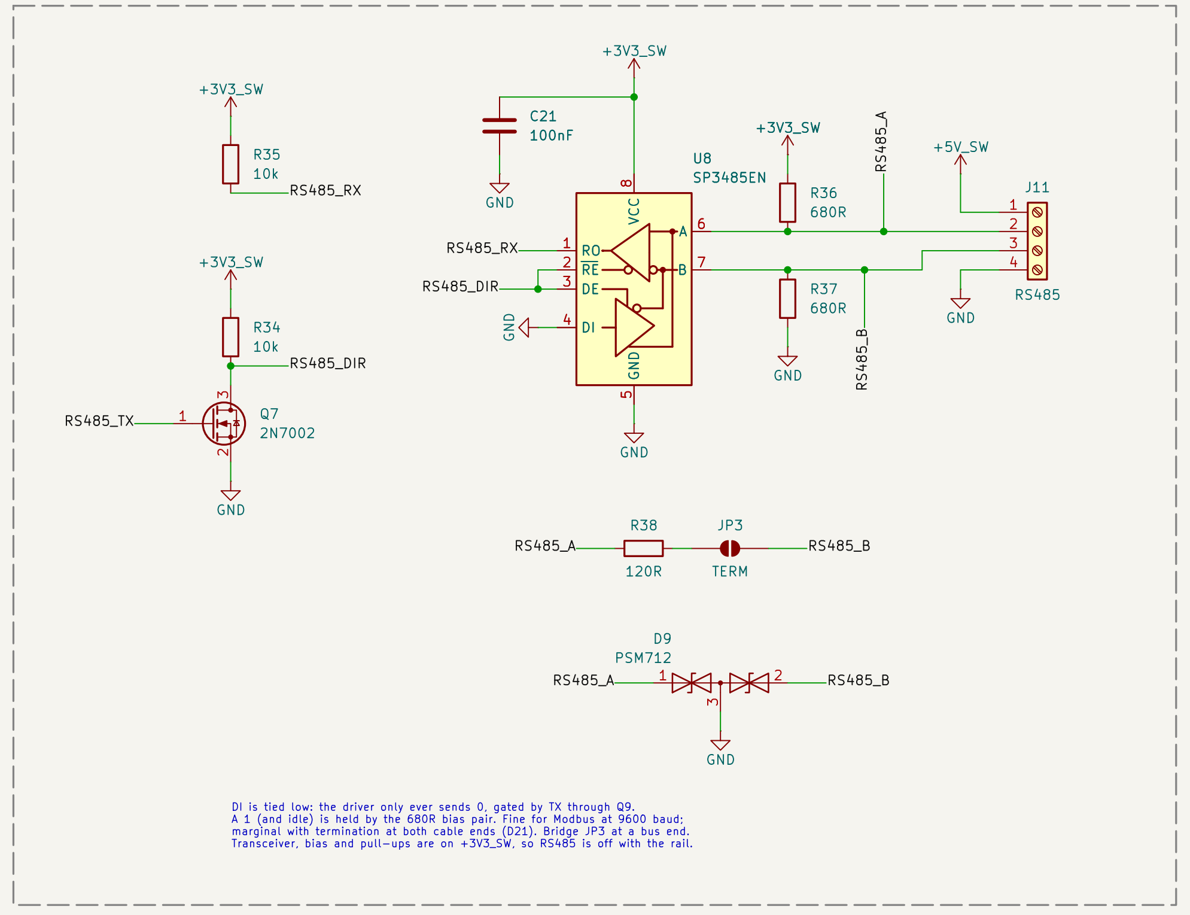
Pads marked [C6]/[C3] differ between the two modules (D15, D18).
Everything essential uses pads that work on both; BTN2 and GPIO_A/B work
only with a C6. SW2 is BOOT and BTN1 (IO9 strap). IO8/IO9 must be high at reset.
C17/C18 close to module pin 3.

SENSOR PORTS: I2C, OLED, GPIO



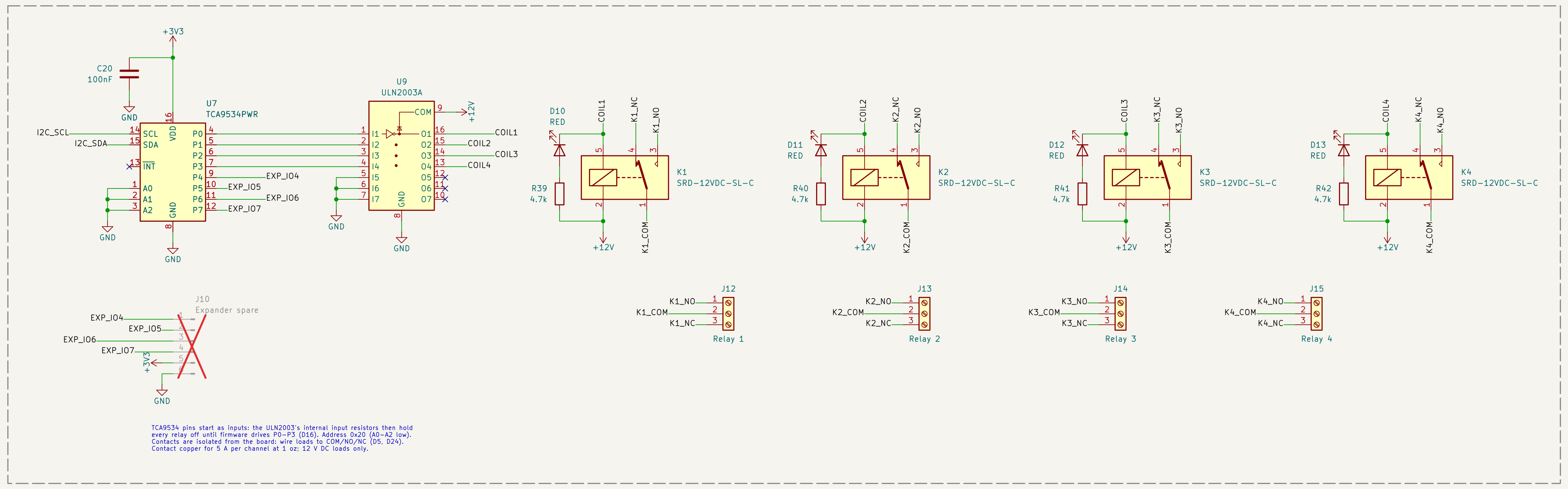
Pull-ups are on +3V3_SW, so nothing back-powers sensors when the rail is off.
Firmware must switch +3V3_SW on before ANY I2C traffic, including relay changes:
the TCA9534 (on always-on 3V3) holds its outputs meanwhile (D20). OLED shares the bus.

RS485 (DIRECTION FROM TX)



DI is tied low: the driver only ever sends 0, gated by TX through Q9.
A 1 (and idle) is held by the 680R bias pair. Fine for Modbus at 9600 baud;
marginal with termination at both cable ends (D21). Bridge JP3 at a bus end.
Transceiver, bias and pull-ups are on +3V3_SW, so RS485 is off with the rail.

RELAY DRIVER & RELAYS



TCA9534 pins start as inputs: the ULN2003's internal input resistors then hold
every relay off until firmware drives P0–P3 (D16). Address 0x20 (A0–A2 low).
Contacts are isolated from the board: wire loads to COM/NO/NC (D5, D24).
Contact copper for 5 A per channel at 1 oz; 12 V DC loads only.

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Natural Innovation
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Title: Frugal-IoT Irrigation V0.1
Size: A1
Date: 2026-10-15
Rev: 0.1
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